

To: ACME CTO

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Subject: Team 9 - Blue Team Report

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1 FTP

A vulnerability that was discovered is that anonymous FTP access was enabled. This allows anyone to access the FTP server with the username anonymous. This led to the exposure of sensitive information such as login information and user access. To resolve this issue, changes were made to the vsftpd.conf file, which holds the configuration for the ftp server. The following changes were made: anonymous_enable was set to no, and no_anon_password was set to no. These changes were made by logging into the ssh server with Mallory's credentials and editing the file that is located at /etc/vsftpd.conf.

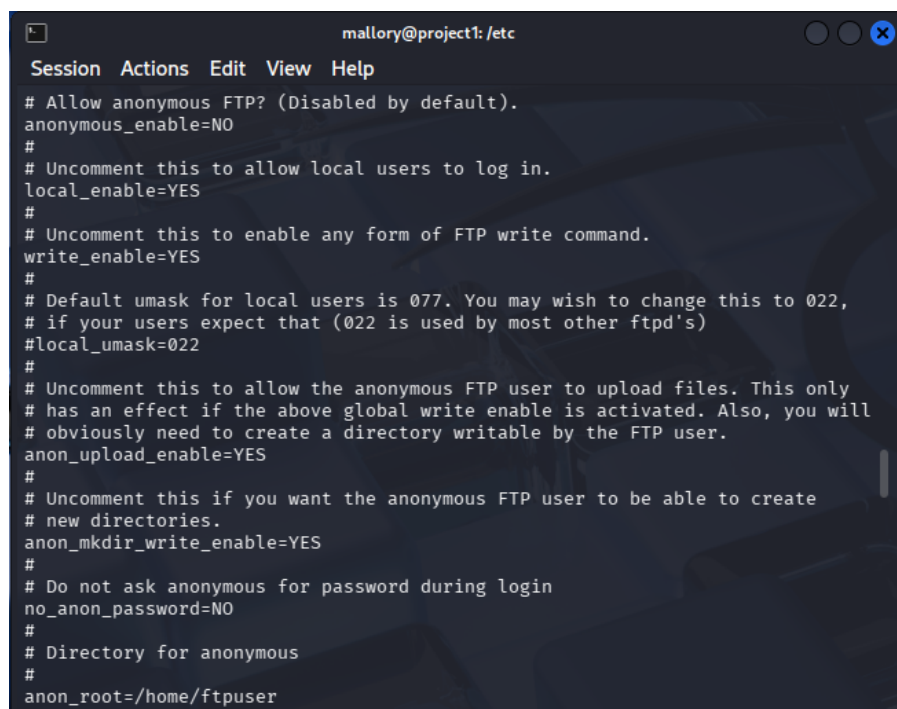


Figure 1: Changes in vsftpd.conf

Listing 1: Changing FTP configuration

```
1 Script started on 2026-02-11 13:33:15-05:00 [TERM="xterm-256color" TTY="/dev/pts/0" COLUMNS="77" LINES="28"]
2 $ ssh 192.168.56.21 mallory@192.168.56.21
3
4 The authenticity of host '192.168.56.21 (192.168.56.21)' can't be established.
5 ED25519 key fingerprint is: SHA256:GLGPoFP82md1DAiuAqu95XVNQeToODKCCXAWG13RPFs
6 This key is not known by any other names.
7 Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
8 Warning: Permanently added '192.168.56.21' (ED25519) to the list of known hosts.
9
10
11 mallory@192.168.56.21's password:
12 Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-94-generic x86_64)
13
```

```

14 * Documentation: https://help.ubuntu.com
15 * Management: https://landscape.canonical.com
16 * Support: https://ubuntu.com/pro
17
18 System information as of Wed Feb 11 06:33:46 PM UTC 2026
19
20 System load: 0.05 Memory usage: 6% Processes: 121
21 Usage of /: 28.4% of 9.75GB Swap usage: 0% Users logged in: 0
22
23
24 Expanded Security Maintenance for Applications is not enabled.
25
26 0 updates can be applied immediately.
27
28 Enable ESM Apps to receive additional future security updates.
29 See https://ubuntu.com/esm or run: sudo pro status
30
31
32 The list of available updates is more than a week old.
33 To check for new updates run: sudo apt update
34 Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
35
36
37 To run a command as administrator (user "root"), use "sudo <command>".
38 See "man sudo_root" for details.
39
40 [?2004h]0;mallory@project1: ~[01;32mmallory@project1[00m:[01;34m~[00m$ cd ..
41 [?2004l
42 [?2004h]0;mallory@project1: /home[01;32mmallory@project1[00m:[01;34m/home[00m$ ls[K[Kcd ..
43 [?2004l
44 [?2004h]0;mallory@project1: /[01;32mmallory@project1[00m:[01;34m/[00m$ ls
45 [?2004l
46 [0m[01;36mbin[0m [01;34metc[0m [01;34mlost+found[0m [01;34mroot[0m [01;34msrv[0
47 [01;34mbin.usr-is-merged[0m [01;34mhome[0m [01;34media[0m [01;34mrun[0m [01;34msys[0m
48 [01;34mboot[0m [01;36mli[b[0m [01;34mmnt[0m [01;36msbin[0m [30;42mtmp[0m
49 [01;34mcdrom[0m [01;36mli[b64[0m [01;34mopt[0m [01;34msbin.usr-is-merged[0m [01;34musr[0m
50 [01;34mdev[0m [01;34mli[b.usr-is-merged[0m [01;34mproc[0m [01;34msnap[0m [01;34mvar[0m
51 [?2004h]0;mallory@project1: /[01;32mmallory@project1[00m:[01;34m/[00m$ cd etc
52 [?2004l
53 [?2004h]0;mallory@project1: /etc[01;32mmallory@project1[00m:[01;34m/etc[00m$ sudo nano vsftpd.conf
54 [?2004l
55 [sudo] password for mallory:
56 [?2004h[?1049h[22;0;0t[1;28r(B[m[41[?7h[39;49m[?1h=[?1h=[?25l[39;49m(B[m[H[2J[26;32H(B[0;7m[ Reading... ](B[m[26;30H(B[0;7m[
57 [27d(B[0;7m^G(B[m Help[27;16H(B[0;7m^O(B[m Write Out (B[0;7m^W(B[m Where Is (B[0;7m^K(B[m Cut[27;61H(B[0;7m^T(B[m
58 Execute
59 [28d(B[0;7m^X(B[m Exit[28;16H(B[0;7m^R(B[m Read File (B[0;7m^\(B[m Replace (B[0;7m^U(B[m Paste[61G(B[0;7m^J(B[m Justify
60 [26d(B[0;7mSave modified buffer?
61 [28d(B[0;7m N(B[m No [28;16H (B[0;7m^C(B[m Cancel[K[26;23H[?12l[?25h[?25l
62 [27d(B[0;7m^G(B[m Help[27;20H(B[0;7mM-D(B[m DOS Format (B[0;7mM-A(B[m Append[27;58H(B[0;7mM-B(B[m Backup File
63 [28d(B[0;7m^C(B[m Cancel[17G (B[0;7mM-M(B[m Mac Format (B[0;7mM-P(B[m Prepend[28;58H(B[0;7m^T(B[m Browse
64 [26d(B[0;7mFile Name to Write: vsftpd.conf(B[m[?12l[?25h[?25l[1K (B[0;7m[ Writing... ](B[m[K[1;48H(B[0;7m [76G(B[m[26;30H(B
65 [J[28d[?12l[?25h[28;1H[?1049l[23;0;0t
66 [?1l>[?2004l[?2004h]0;mallory@project1: /etc[01;32mmallory@project1[00m:[01;34m/etc[00m$ stop
67 [?2004l
68 Command 'stop' not found, but there are 20 similar ones.
69 [?2004h]0;mallory@project1: /etc[01;32mmallory@project1[00m:[01;34m/etc[00m$ exit
70 [?2004l
71 logout
72 Connection to 192.168.56.21 closed.
73
74 $ stop
75 sh: 2: stop: not found
76 $ end
77 sh: 3: end: not found
78 $ exit
79
80 Script done on 2026-02-11 13:36:31-05:00 [COMMAND_EXIT_CODE="127"]

```

After a quick reboot of the system, the changes were put into place, and the anonymous login was not able to have access to the ftp server. This makes the ftp server much more secure because now only users with credentials are able to log in and utilize the ftp server.

Listing 2: Anonymous Login Failure

```

1 Script started on 2026-02-11 14:01:45-05:00 [TERM="xterm-256color" TTY="/dev/pts/0" COLUMNS="77" LINES="28"]
2 $ ftp op
3 ftp> open 192.168.56.21
4 Connected to 192.168.56.21.
5 220 (vsFTPd 3.0.5)
6 Name (192.168.56.21:Kate): anonymous
7 331 Please specify the password.
8 Password:
9 530 Login incorrect.
10 ftp: Login failed
11 ftp> exit
12 221 Goodbye.
13 $ exit
14
15 Script done on 2026-02-11 14:02:03-05:00 [COMMAND_EXIT_CODE="0"]

```

One file under the ftpuser directory contained a password for alice. To remediate this error,

we deleted the file from the system entirely.

```
mallory@project1:~$ cat /home/ftpuser/mallory/notes/note_4.txt
== INTERNAL IT NOTE ==
User: alice
Temp password reset during migration

username: alice
password: nicole

Reminder:
User was supposed to change this after first login.
Follow up if access issues persist.
mallory@project1:~$ rm /home/ftpuser/mallory/notes/note_4.txt
rm: remove write-protected regular file '/home/ftpuser/mallory/notes/note_4.t
xt'? y
rm: cannot remove '/home/ftpuser/mallory/notes/note_4.txt': Permission denied
mallory@project1:~$ sudo rm /home/ftpuser/mallory/notes/note_4.txt
[sudo] password for mallory:
mallory@project1:~$ cat /home/ftpuser/mallory/notes/note_4.txt
cat: /home/ftpuser/mallory/notes/note_4.txt: No such file or directory
mallory@project1:~$
```

Figure 2: Delete File Containing Password

2 Password Policies

Passwords are a key part of any modern secure system. They provide employees and technicians convenient access to remote systems without the need for direct physical access or moving between locations. The use of passwords does introduce some risk, however, this risk can be mostly mitigated with proper system and administrative policies.

2.1 System Policies

Proper password complexity requirements are crucial to maintaining a secure system. It should be at least twelve characters long and contain lowercase letters, capital letters, numbers, and symbols. An ideal password should also exclude any Personally Identifiable Information like your birthday, relative name, pet names, or hobbies. Length, complexity, and a lack of direct association can drastically reduce the risk of brute force or socially engineered guesses.

We added a password length requirement of 12 to the /pam.d/common-password configuration file. Complexity enforcement requires an additional package from the internet, so while we did not enable that, it would be trivial to install pam_pwquality.so and add the password complexity requirements to the configuration as well.

As stated in our red team report, we discovered multiple users with overly simple passwords: alice and bob. Both users had passwords consisting of a single, simple, word. To mitigate the immediate risk both users were assigned, 16 character, randomly generated passwords containing a mixture of capital letters, lowercase letters, symbols, and numbers. These passwords were marked as expired and should require the users to reset them on next login. These two changes should heavily reduce the risk faced due to insecure passwords.

Listing 3: Implementing Password Policy Corrections

```
1 Script started on 2026-02-12 15:44:28-05:00 [TERM="xterm-256color" TTY="/dev/pts/0" COLUMNS="77" LINES="28"]
2 ]0;kali@kali: ~/scripts[J(kalikal)-[~/scripts]
3 $ [?1h=[?2004hexitexitftp 192.168.56.21[15Dp [13C[17Dssh mallory@192.168.56.21[25Dexit [21Ditssh
4 mallory@192.168.56.21[23Dh [21C[?1l>[?2004l
5 The authenticity of host '192.168.56.21 (192.168.56.21)' can't be established.
6 ED25519 key fingerprint is: SHA256:GLGPoFP82md1DAiuAqu95XVNQeToODKCCXAWG13RPFs
7 This key is not known by any other names.
8 Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
9 Warning: Permanently added '192.168.56.21' (ED25519) to the list of known hosts.
10 mallory@192.168.56.21's password:
Permission denied, please try again.
```

[illegible]

```

75 | sync:x:4:65534:sync:/bin:/bin/sync
76 | games:x:5:60:games:/usr/games:/usr/sbin/nologin
77 | man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
78 | lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
79 | mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
80 | news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
81 | uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
82 | proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
83 | www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
84 | backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
85 | list:x:38:38:Mailng List Manager:/var/list:/usr/sbin/nologin
86 | irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
87 | _apt:x:42:65534::/nonexistent:/usr/sbin/nologin
88 | nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
89 | systemd-network:x:998:998:systemd Network Management:/usr/sbin/nologin
90 | systemd-timesync:x:997:997:systemd Time Synchronization:/usr/sbin/nologin
91 | dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
92 | messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
93 | systemd-resolve:x:992:992:systemd Resolver:/usr/sbin/nologin
94 | pollinate:x:102:1::/var/cache/pollinate:/bin/false
95 | polkitd:x:991:991:User for polkitd:/usr/sbin/nologin
96 | syslog:x:103:104::/nonexistent:/usr/sbin/nologin
97 | uidd:x:104:105::/run/uidd:/usr/sbin/nologin
98 | tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
99 | tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
100 | landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
101 | fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
102 | usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
103 | sshd:x:109:65534::/run/ssh:/usr/sbin/nologin
104 | project1:x:1000:1000:project1:/home/project1:/bin/bash
105 | vboxadd:x:999:1::/var/run/vboxadd:/bin/false
106 | alice:x:1001:100::/home/alice:/bin/bash
107 | bob:x:1002:100::/home/bob:/bin/bash
108 | eve:x:1003:100::/home/eve:/bin/bash
109 | mallory:x:1004:100::/home/mallory:/bin/bash
110 | ftp:x:110:111:ftp daemon,,,:/srv/ftp:/usr/sbin/nologin
111 | ftpuser:x:1005:100::/home/ftpuser:/bin/bash
112 | [?2004h]0;mallory@project1: ~mallory@project1:~$ sudo passwd alice
113 | [?2004lNew password:
114 | Retype new password:
115 | passwd: password updated successfully
116 | [?2004h]0;mallory@project1: ~mallory@project1:~$ sudo passwd bob
117 | [?2004lNew password:
118 | Retype new password:
119 | passwd: password updated successfully
120 | [?2004h]0;mallory@project1: ~mallory@project1:~$ sudo passwd -e alice
121 | passwd: password changed
122 | [?2004h]0;mallory@project1: ~mallory@project1:~$ sudo passwd -e bob
123 | passwd: password changed
124 | [?2004h]0;mallory@project1: ~mallory@project1:~$ exit
125 | [?2004llogout
126 | Connection to 192.168.56.21 closed.
127 |
128 | [J(kalikali)-[~/scripts] ]0;kali@kali: ~/scripts
129 | $ [?1h=[?2004heee[38;5;244mxitexexxiexit[?1l>[?2004l
130 |
131 | Script done on 2026-02-12 15:54:39-05:00 [COMMAND_EXIT_CODE="0"]

```

2.2 Administrative Policies

Another major factor impacting password security in our red team discoveries were files containing passwords. This uncovers a need for additional corporate training and policies regarding password security.

1. **Password Storage** - Passwords should NEVER be stored in plain text or in a format trivial to decode. If a file containing passwords is necessary for business use-cases, these files should at minimum be encrypted using a secure password.
2. **Passwords Containing Personal Information** - Passwords should NEVER include personal information. This is inherently challenging to police from system policies, so additional training and policies should restrict this behavior in the management plane (corporate policies).
3. **Password Expiration** - While it is no longer best practice to force users to frequently rotate passwords, users should be required to reset their password on first login. This can be done on-device, but additional corporate policies need to be in place.

```
(student@kali)-[~/Downloads]
$ cat users.bak
# Legacy migration backup (DO NOT EXPOSE)

bob
temporary password: charlie

alice
password reset pending

mallory
access unchanged

(mallory@project1:/home/ftpuser/mallory/notes$ cat *.txt
Ticket: #1769782698
Reminder: Do not share credentials.
Ticket: #1769782698
Reminder: Do not share credentials.
Ticket: #1769782698
Reminder: Do not share credentials.
== INTERNAL IT NOTE ==
User: alice
Temp password reset during migration

username: alice
password: nicole

Reminder:
User was supposed to change this after first login.
Follow up if access issues persist.
mallory@project1:/home/ftpuser/mallory/notes$
```

Figure 3: Bob and Alice Passwords in Files

3 Web Server

Sensitive information was exposed through the Apache web server, including internal and backup content that should not have been publicly accessible. This exposure increases the risk of credential disclosure and follow-on compromise. The web service was kept in place, but hardened by removing sensitive content from the web root, disabling directory listing, blocking access to common sensitive file types, and correcting file permissions.

3.1 Remove Sensitive Files from Web Root

Sensitive directories identified during the Red Team assessment were moved out of `/var/www/html` and stored in a protected directory owned by `root`. This preserves the files for administrative use while preventing public access.

Listing 4: Commands used to remove sensitive directories from the web root

```
sudo mkdir -p /var/secure_archive
sudo chmod 700 /var/secure_archive
sudo chown root:root /var/secure_archive
```

```
sudo mv /var/www/html/backup /var/secure_archive/  
sudo mv /var/www/html/internal /var/secure_archive/  
  
# Before/after verification  
ls -la /var/www/html  
sudo ls -la /var/secure_archive
```

3.2 Disable Directory Listing

Apache directory indexing was disabled to prevent attackers from enumerating directory contents when no index file is present.

Listing 5: Commands used to disable Apache directory listing

```
# Edit Apache configuration to ensure:  
# Options -Indexes  
sudo nano /etc/apache2/apache2.conf  
  
# Validate config and apply changes  
sudo apache2ctl configtest  
sudo systemctl restart apache2
```

3.3 Block Access to Sensitive File Types

Apache was configured to deny access to common sensitive file extensions (e.g., backups and configuration files). This provides defense-in-depth if a sensitive file is accidentally placed in a web-accessible directory in the future.

Listing 6: Commands used to block access to sensitive file extensions

```
sudo nano /etc/apache2/conf-available/deny-sensitive.conf  
sudo a2enconf deny-sensitive  
  
sudo apache2ctl configtest  
sudo systemctl reload apache2
```

3.4 Verification

Configuration integrity was verified using `apache2ctl configtest`, and the Apache service was restarted/reloaded to apply changes.

Listing 7: Verification commands confirming sensitive paths are no longer exposed

```
curl -I http://localhost/backup  
curl -I http://localhost/internal  
  
systemctl status apache2 --no-pager
```

The full command transcript is provided in Appendix 12.

4 Home Directories

4.1 Initial Assessment

The red team successfully enumerated user home directories due to overly permissive access controls (755/750). This configuration allowed lateral movement opportunities where any authenticated attacker could:

- Enumerate user data and sensitive documents
- Identify targets for privilege escalation
- Map organizational structure through file analysis
- Harvest credentials or SSH keys if improperly secured

This represents a critical information disclosure vulnerability and violates defense-in-depth principles. The initial assessment revealed that alice, bob, eve, and mallory had '750' permissions: group readable/executable, and ftpuser had '755' permissions: world readable/executable.

Listing 8: Home Directory Permissions

```
1 Script started on 2026-02-12 15:29:56+00:00 [TERM="linux" TTY="/dev/tty1" COLUMNS="160" LINES="50"]
2 [?2004hmallory@project1:~/blue-work$ grep -Ri "password" /home/cd ..[Krm webserver.txt [4Pcd blue-work/grep -Ri "password" /
   home/[2Pfind /home/ f -perm -004[10Pls -ld /home/*
3 [?2004l
4 drwxr-x--- 7 alice users 4096 Feb 3 17:17 [0m[01;34m/home/alice[0m
5 drwxr-x--- 9 bob users 4096 Jan 30 19:03 [01;34m/home/bob[0m
6 drwxr-x--- 6 eve users 4096 Jan 30 14:18 [01;34m/home/eve[0m
7 drwxr-xr-x 7 ftpuser users 4096 Jan 30 19:13 [01;34m/home/ftpuser[0m
8 drwxr-x--- 10 mallory users 4096 Feb 12 03:09 [01;34m/home/mallory[0m
9 drwxr-x--- 4 project1 project1 4096 Feb 1 23:22 [01;34m/home/project1[0m
```

4.2 Filesystem Permissions Hardening

We addressed this by hardening filesystem permissions to prevent unauthorized enumeration and lateral movement. Applied principle of least privilege by restricting home directory access to owner only, giving Alice, Bob, Eve, Mallory, and ftpuser '700' permissions: owner only.

Listing 9: Change Permissions Commands

```
1 [?2004hmallory@project1:~/blue-work$ sudo chmod 7-- [K[K[K00 /home/f[Kt[Kftuser/
2 [?2004l
3 [sudo] password for mallory:
4 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/ftpuser/[C[C[C[K[K[K[K[K[K[K[Kalice/
5 [?2004l
6 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/alice/[K[K[K[K[K[K[Kbob/
7 [?2004l
8 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/bob/alice/[2Pbob/[K[K[K[Keve/
9 [?2004l
10 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/eve/[K[K[K[Kmallory/
11 [?2004l
12 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/mallory/[4Peve/bob/alice/ftpuser/[15Pls -ld /home/*grep -Ri "
   password" /home/cd ..[Kgrep -Ri "password" /home/[12Pls -ld /home/*
13 [?2004l
```

4.3 Post-Remediation Validation

The Post-remediation permission audit confirmed that all user directories were now properly secured.

Listing 10: Confirmation of Permissions

```
1 drwx----- 7 alice users 4096 Feb 3 17:17 [0m[01;34m/home/alice[0m
2 drwx----- 9 bob users 4096 Jan 30 19:03 [01;34m/home/bob[0m
3 drwx----- 6 eve users 4096 Jan 30 14:18 [01;34m/home/eve[0m
4 drwx----- 7 ftpuser users 4096 Jan 30 19:13 [01;34m/home/ftpuser[0m
5 drwx----- 10 mallory users 4096 Feb 12 03:09 [01;34m/home/mallory[0m
6 drwxr-x--- 4 project1 project1 4096 Feb 1 23:22 [01;34m/home/project1[0m
```

All home directories now implement 700 permissions (owner-only access). Lateral movement via filesystem enumeration has been effectively blocked. Threat actors with initial access can no longer pivot to other user contexts through directory traversal.

4.4 Defense-in-Depth Validation

Next, we conducted a comprehensive filesystem audit to identify residual misconfigurations that could enable data exfiltration or privilege escalation. This revealed zero unauthorized file exposures across all user directories, individual file permissions confirmed restrictive (640 or stronger), no world-writable files identified, and no residual attack surface identified in filesystem layer.

Listing 11: Scan for Word-Readable/Writeable Files or Directories

```
1 [?2004hmallory@project1:~/blue-work$ find /hm[Kome/[K -type [K[K[K[K[K[K[K/[K =[K-type f -perm -004 2>/dev/null [K
2 [?2004hmallory@project1:~/blue-work$ find /home -type f -perm -004 2>/dev/null[C[1P 2>/dev/null2 2>/dev/null
3 [?2004l
4 [?2004hmallory@project1:~/blue-work$ find /home -type f -perm -002 2>/dev/null[1P[1@d[C[C[C[C[C[C[C[C[C[C[C[1P 2>/dev/null5
5 2>/dev/null
[?2004l
```

The full command transcript is provided in Appendix 13.

A Command Transcripts

A.1 Web Server Remediation Transcript

Listing 12: Web Server remediation transcript captured using script

```
1 Script started on 2026-02-12 06:45:30+00:00 [TERM="linux" TTY="/dev/tty1" COLUMNS="160" LINES="50"]
2 To run a command as administrator (user "root"), use "sudo <command>".
3 See "man sudo_root" for details.
4
5 [?2004hmallory@project1:~$ whoami
6 [?2004l
7 mallory
8 [?2004hmallory@project1:~$ sudo -v
9 [?2004l
10 [sudo] password for mallory:
11 [?2004hmallory@project1:~$ systemctl status apache2 --no-pager
12 [?2004l
13     apache2.service - The Apache HTTP Server
14     Loaded: loaded (/usr/lib/systemd/system/apache2.service; vendor preset: enabled)
15     Active: active (running) since Thu 2026-02-12 06:38:23 UTC; 8min ago
16     Docs: https://httpd.apache.org/docs/2.4/
17     Process: 756 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
18     Main PID: 828 (apache2)
19     Tasks: 55 (limit: 4603)
20     Memory: 7.7M (peak: 8.0M)
21     CPU: 982ms
22     CGroup: /system.slice/apache2.service
23             └─828 /usr/sbin/apache2 -k start
24             └─839 /usr/sbin/apache2 -k start
25             └─841 /usr/sbin/apache2 -k start
26
27 Warning: some journal files were not opened due to insufficient permissions.
28 [?2004hmallory@project1:~$ ls -la /var/www/html
29 [?2004l
30 total 48
31 drwxr-xr-x 7 www-data www-data 4096 Jan 30 17:34 .
32 drwxr-xr-x 3 root root 4096 Jan 30 17:22 ..
33 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:29 backup
34 -rwxr-xr-x 1 root root 10671 Jan 30 17:26 index-default.html
35 -rwxr-xr-x 1 www-data www-data 220 Jan 30 17:27 index.html
36 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:36 internal
37 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:35 login
38 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:24 notes
39 -rw-r--r-- 1 root root 54 Jan 30 17:28 robots.txt
40 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:24 uploads
41 [?2004hmallory@project1:~$ sudo mkdir -p /var/secure_archive
42 [?2004l
43 [?2004hmallory@project1:~$ sudo chmod 700 /var/secure_archive
44 [?2004l
45 [?2004hmallory@project1:~$ sudo chown root:root /var/secure_archive
46 [?2004l
47 [?2004hmallory@project1:~$ ls -ld /var/secure_archive
48 [?2004l
49 drwx----- 2 root root 4096 Feb 12 06:47 /var/secure_archive
50 [?2004hmallory@project1:~$ sudo mv /var/www/html/backup /var/secure_archive/ 2>/dev/null
51 [?2004l
52 [?2004hmallory@project1:~$ sudo mv /var/www/html/internal /var/secure_archive/ 2>/dev/null
53 [?2004l
54 [?2004hmallory@project1:~$ ls -la /var/www/html
55 [?2004l
56 total 40
57 drwxr-xr-x 5 www-data www-data 4096 Feb 12 06:50 .
58 drwxr-xr-x 3 root root 4096 Jan 30 17:22 ..
59 -rwxr-xr-x 1 root root 10671 Jan 30 17:26 index-default.html
60 -rwxr-xr-x 1 www-data www-data 220 Jan 30 17:27 index.html
```

```

61 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:35 login
62 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:24 notes
63 -rw-r--r-- 1 root root 54 Jan 30 17:28 robots.txt
64 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:24 uploads
65 [?2004hmallory@project1:~$ ls -la /var/secure_archive
66 [?2004l
67 ls: cannot open directory '/var/secure_archive': Permission denied
68 [?2004hmallory@project1:~$ sudo ls -la /var/secure_archive
69 [?2004l
70 total 16
71 drwx----- 4 root root 4096 Feb 12 06:50 .
72 drwxr-xr-x 15 root root 4096 Feb 12 06:47 ..
73 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:29 backup
74 drwxr-xr-x 2 www-data www-data 4096 Jan 30 17:36 internal
75 [?2004hmallory@project1:~$ sudo nano /etc/apache2/apache2.conf
76 [?2004l
77 [?2004h)0[?7h[?25l[?1c[ Reading... ][ Read 225 lines ] GNU nano 7.2 /
    etc/apache2/apache2.conf
78 ^G Help^O Write Out ^W Where Is ^K Cut^T Execute^C Location M-U UndoM-A Set Mark M-] To Bracket M-Q Previous
79 ^X Exit^R Read File ^\ Replace^U Paste^J Justify^/ Go To Line M-E RedoM-6 Copy^Q Where Was M-W Next
80 # This is the main Apache server configuration file. It contains the
81 # configuration directives that give the server its instructions.
82 # See http://httpd.apache.org/docs/2.4/ for detailed information about
83 # the directives and /usr/share/doc/apache2/README.Debian about Debian specific
84 # hints.
85 ### Summary of how the Apache 2 configuration works in Debian:
86 # The Apache 2 web server configuration in Debian is quite different to
87 # upstream's suggested way to configure the web server. This is because Debian's
88 # default Apache2 installation attempts to make adding and removing modules,
89 # virtual hosts, and extra configuration directives as flexible as possible, in
90 # order to make automating the changes and administering the server as easy as
91 # possible.
92 # It is split into several files forming the configuration hierarchy outlined
93 # below, all located in the /etc/apache2/ directory:
94 ##/etc/apache2/
95 #|-- apache2.conf
96 #| |-- ports.conf
97 #| |-- mods-enabled
98 #| |-- *.load
99 #| |-- *.conf
100 #| |-- conf-enabled
101 #| |-- *.conf
102 # |-- sites-enabled
103 # |-- *.conf
104 ### * apache2.conf is the main configuration file (this file). It puts the pieces
105 # together by including all remaining configuration files when starting up the
106 # web server.
107 ## * ports.conf is always included from the main configuration file. It is
108 # supposed to determine listening ports for incoming connections which can be
109 # customized anytime.
110 ## * Configuration files in the mods-enabled/, conf-enabled/ and sites-enabled/
111 # directories contain particular configuration snippets which manage modules,
112 # global configuration fragments, or virtual host configurations,
113 # respectively.
114 ## They are activated by symlinking available configuration files from their
115 # respective *-available/ counterparts. These should be managed by using our
116 # helpers a2enmod/a2dismod, a2ensite/a2dissite and a2enconf/a2disconf. See
117 # their respective man pages for detailed information.
118 [?25l[?1c[?25h[?0c78
119 [?25l[?1c[?25h[?0c78
120 # * The binary is called apache2. Due to the use of environment variables, in
121 [?25l[?1c[?25h[?0c78
122 # the default configuration, apache2 needs to be started/stopped with
123 [?25l[?1c[?25h[?0c78
124 # /etc/init.d/apache2 or apache2ctl. Calling /usr/bin/apache2 directly will not
125 [?25l[?1c[?25h[?0c78
126 # work with the default configuration.
127 [?25l[?1c[?25h[?0c78
128 [?25l[?1c[?25h[?0c78
129 [?25l[?1c[?25h[?0c78
130 # Global configuration
131 [?25l[?1c[?25h[?0c78
132 [?25l[?1c[?25h[?0c78
133 [?25l[?1c[?25h[?0c78
134 [?25l[?1c[?25h[?0c78
135 # ServerRoot: The top of the directory tree under which the server's
136 [?25l[?1c[?25h[?0c78
137 # configuration, error, and log files are kept.
138 [?25l[?1c[?25h[?0c78
139 [?25l[?1c[?25h[?0c78
140 # NOTE! If you intend to place this on an NFS (or otherwise network)
141 [?25l[?1c[?25h[?0c78
142 # mounted filesystem then please read the Mutex documentation (available
143 [?25l[?1c[?25h[?0c78
144 # at <URL:http://httpd.apache.org/docs/2.4/mod/core.html#mutex>);
145 [?25l[?1c[?25h[?0c78
146 # you will save yourself a lot of trouble.
147 [?25l[?1c[?25h[?0c78
148 [?25l[?1c[?25h[?0c78
149 # Do NOT add a slash at the end of the directory path.
150 [?25l[?1c[?25h[?0c78
151 [?25l[?1c[?25h[?0c78
152 #ServerRoot "/etc/apache2"
153 [?25l[?1c[?25h[?0c78
154 [?25l[?1c[?25h[?0c78
155 [?25l[?1c[?25h[?0c78
156 # The accept serialization lock file MUST BE STORED ON A LOCAL DISK.
157 [?25l[?1c[?25h[?0c78
158 [?25l[?1c[?25h[?0c78

```

```

159 #Mutex file:${APACHE_LOCK_DIR} default
160 [?251[?1c[?25h[?0c78
161 [?251[?1c[?25h[?0c78
162 #[?251[?1c[?25h[?0c78
163 # The directory where shm and other runtime files will be stored.
164 [?251[?1c[?25h[?0c78
165 #[?251[?1c[?25h[?0c78
166 [?251[?1c[?25h[?0c78
167 DefaultRuntimeDir ${APACHE_RUN_DIR}
168 [?251[?1c[?25h[?0c78
169 [?251[?1c[?25h[?0c78
170 #[?251[?1c[?25h[?0c78
171 # PidFile: The file in which the server should record its process
172 [?251[?1c[?25h[?0c78
173 # identification number when it starts.
174 [?251[?1c[?25h[?0c78
175 # This needs to be set in /etc/apache2/envvars
176 [?251[?1c[?25h[?0c78
177 #[?251[?1c[?25h[?0c78
178 PidFile ${APACHE_PID_FILE}
179 [?251[?1c[?25h[?0c78
180 [?251[?1c[?25h[?0c78
181 #[?251[?1c[?25h[?0c78
182 # Timeout: The number of seconds before receives and sends time out.
183 [?251[?1c[?25h[?0c78
184 #[?251[?1c[?25h[?0c78
185 Timeout 300
186 [?251[?1c[?25h[?0c78
187 [?251[?1c[?25h[?0c78
188 #[?251[?1c[?25h[?0c78
189 # KeepAlive: Whether or not to allow persistent connections (more than
190 [?251[?1c[?25h[?0c78
191 # one request per connection). Set to "Off" to deactivate.
192 [?251[?1c[?25h[?0c78
193 #[?251[?1c[?25h[?0c78
194 KeepAlive On
195 [?251[?1c[?25h[?0c78
196 [?251[?1c[?25h[?0c78
197 #[?251[?1c[?25h[?0c78
198 # MaxKeepAliveRequests: The maximum number of requests to allow
199 [?251[?1c[?25h[?0c78
200 # during a persistent connection. Set to 0 to allow an unlimited amount.
201 [?251[?1c[?25h[?0c78
202 # We recommend you leave this number high, for maximum performance.
203 [?251[?1c[?25h[?0c78
204 #[?251[?1c[?25h[?0c78
205 MaxKeepAliveRequests 100
206 [?251[?1c[?25h[?0c78
207 [?251[?1c[?25h[?0c78
208 #[?251[?1c[?25h[?0c78
209 # KeepAliveTimeout: Number of seconds to wait for the next request from the
210 [?251[?1c[?25h[?0c78
211 # same client on the same connection.
212 [?251[?1c[?25h[?0c78
213 #[?251[?1c[?25h[?0c78
214 KeepAliveTimeout 5
215 [?251[?1c[?25h[?0c78
216 [?251[?1c[?25h[?0c78
217 [?251[?1c[?25h[?0c78
218 # These need to be set in /etc/apache2/envvars
219 [?251[?1c[?25h[?0c78
220 User ${APACHE_RUN_USER}
221 [?251[?1c[?25h[?0c78
222 Group ${APACHE_RUN_GROUP}
223 [?251[?1c[?25h[?0c78
224 [?251[?1c[?25h[?0c78
225 #[?251[?1c[?25h[?0c78
226 # HostnameLookups: Log the names of clients or just their IP addresses
227 [?251[?1c[?25h[?0c78
228 # e.g., www.apache.org (on) or 204.62.129.132 (off).
229 [?251[?1c[?25h[?0c78
230 # The default is off because it'd be overall better for the net if people
231 [?251[?1c[?25h[?0c78
232 # had to knowingly turn this feature on, since enabling it means that
233 [?251[?1c[?25h[?0c78
234 # each client request will result in AT LEAST one lookup request to the
235 [?251[?1c[?25h[?0c78
236 # nameserver.
237 [?251[?1c[?25h[?0c78
238 #[?251[?1c[?25h[?0c78
239 HostnameLookups Off
240 [?251[?1c[?25h[?0c78
241 [?251[?1c[?25h[?0c78
242 # ErrorLog: The location of the error log file.
243 [?251[?1c[?25h[?0c78
244 # If you do not specify an ErrorLog directive within a <VirtualHost>
245 [?251[?1c[?25h[?0c78
246 # container, error messages relating to that virtual host will be
247 [?251[?1c[?25h[?0c78
248 # logged here. If you *do* define an error logfile for a <VirtualHost>
249 [?251[?1c[?25h[?0c78
250 # container, that host's errors will be logged there and not here.
251 [?251[?1c[?25h[?0c78
252 #[?251[?1c[?25h[?0c78
253 ErrorLog ${APACHE_LOG_DIR}/error.log
254 [?251[?1c[?25h[?0c78
255 [?251[?1c[?25h[?0c78
256 #[?251[?1c[?25h[?0c78
257 # LogLevel: Control the severity of messages logged to the error_log.

```



```

339 AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName
    ' directive globally to suppress this message
340 Syntax OK
341 [?2004hmallory@project1:~$ sudo systemctl restart apache2
342 [?2004l
343 [?2004hmallory@project1:~$ sudo nano /etc/apache2/conf-available/deny-sensitive.conf
344 [?2004l
345 [?2004h)0[?7h[?25l[?1c[ New File ] GNU nano 7.2 /etc/apache2/conf-available/deny
    -sensitive.conf
346 ^G Help^O Write Out ^W Where Is ^K Cut^T Execute^C Location M-U UndoM-A Set Mark M-] To Bracket M-Q Previous
347 ^X Exit^R Read File ^\ Replace^U Paste^J Justify^/ Go To Line M-E RedoM-6 Copy^Q Where Was M-W Next
348 [?25h[?0c[?25l[?1c[?25h[?0c
349 <[?25l[?1c[?25h[?0cF[?25l[?1c[?25h[?0ci[?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0ce[?25l[?1c[?25h[?0cs[?25l[?1c[?25h[?0cm[?25l[?1c
    [?25h[?0ca[?25l[?1c[?25h[?0ct[?25l[?1c[?25h[?0cc[?25l[?1c[?25h[?0ch[?25l[?1c[?25h[?0c [?25l[?1c[?25h[?0c" [?25l[?1c[?25h
    [?0c([?25l[?1c[?25h[?0c? [?25l[?1c[?25h[?0ci[?25l[?1c[?25h[?0c) [?25l[?1c[?25h[?0c\ [?25l[?1c[?25h[?0c. [?25l[?1c
350 [?25h[?0c([?25l[?1c[?25h[?0cb[?25l[?1c[?25h[?0ca[?25l[?1c[?25h[?0ck[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0co[?25l[?1c[?25h[?0cl
    [?25l[?1c[?25h[?0cd[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0co[?25l[?1c[?25h[?0cr[?25l[?1c[?25h[?0ci[?25l[?1c[?25h[?0cg[?25l
    [?1c[?25h[?0c] [?25l[?1c[?25h[?0cs[?25l[?1c[?25h[?0cq[?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0cs[?25l[?1c
    [?25h[?0cq[?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0ci[?25l[?1c[?25h[?0ct[?25l[?1c[?25h[?0ce[?25l[?1c[?25h[?0c] [?25l[?1c[?25h
    [?0cd[?25l[?1c[?25h[?0cb[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0ct[?25l[?1c[?25h[?0ca[?25l[?1c[?25h[?0cr[?25l[?1c[?25h[?0c
    [?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0cg[?25l[?1c[?25h[?0cz[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0cz[?25l[?1c[?25h[?0ci[?25l
    [?1c[?25h[?0cp[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0co[?25l[?1c[?25h[?0cg[?25l[?1c[?25h[?0c] [?25l[?1c
    [?25h[?0cc[?25l[?1c[?25h[?0co[?25l[?1c[?25h[?0cn[?25l[?1c[?25h[?0cf[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0ci[?25l[?1c[?25h
    [?0cn[?25l[?1c[?25h[?0ci[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0cy[?25l[?1c[?25h[?0cm[?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0c
    [?25l[?1c[?25h[?0cy[?25l[?1c[?25h[?0ca[?25l[?1c[?25h[?0cm[?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0ce[?25l
    [?1c[?25h[?0cn[?25l[?1c[?25h[?0cv[?25l[?1c[?25h[?0c] [?25l[?1c[?25h[?0ct[?25l[?1c[?25h[?0cx[?25l[?1c[?25h[?0ct[?25l[?1c
    [?25h[?0c] [?25l[?1c[?25h[?0c$[?25l[?1c[?25h[?0c" [?25l[?1c[?25h[?0c>[?25l[?1c
351 [?25h[?0c[?25l[?1c[?25h[?0c[?25l[?1c[?25h[?0cR[?25l[?1c[?25h[?0ce[?25l[?1c[?25h[?0cq[?25l[?1c[?25h[?0cu[?25l[?1c[?25h[?0ci
    [?25l[?1c[?25h[?0cr[?25l[?1c[?25h[?0ce[?25l[?1c[?25h[?0c [?25l[?1c[?25h[?0ca[?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0cl[?25l
    [?1c[?25h[?0c [?25l[?1c[?25h[?0cd[?25l[?1c[?25h[?0cd[?25l[?1c[?25h[?0c [?25l[?1c[?25h[?0ce[?25l[?1c[?25h[?0cn[?25l[?1c
    [?25h[?0cl[?25l[?1c[?25h[?0ce[?25l[?1c[?25h[?0cd[?25l[?1c
352 [?25h[?0c[?25l[?1c[?25h[?0c<[?25l[?1c[?25h[?0c/ [?25l[?1c[?25h[?0ci[?25l[?1c[?25h[?0cl[?25l[?1c[?25h[?0ce
    [?25l[?1c[?25h[?0cs[?25l[?1c[?25h[?0cm[?25l[?1c[?25h[?0ca[?25l[?1c[?25h[?0ct[?25l[?1c[?25h[?0cc[?25l[?1c[?25h[?0ch[?25l
    [?1c[?25h[?0c>[?25l[?1cM-D DOS Forma M-A Append M-B Backup FileC CancelM-M Mac Format M-P
    Prepend ^T Browse
353 File Name to Write: /etc/apache2/conf-available/deny-sensitive.conf [?25h[?0c[?25l[?1c [ Writing
    ... ] [ Wrote 3 lines ] ^O Write Out ^W Where Is ^K Cu^T Execute^C Location M-U UndoM-A Set Mark M-] To
    Bracket M-Q PreviousX Exit ^R Read File ^\ Replace ^U Paste^J Justify^/ Go To Line M-E RedoM-6 Copy ^
    Q Where Was M-W Next[?25h[?0c[?25l[?1c
354 [?25h[?0c
355 [?2004l[?2004hmallory@project1:~$ sudo a2enconf deny-sensitive
356 [?2004l
357 sudo: a2enconf: command not found
358 [?2004hmallory@project1:~$ sudo a2enconf deny-sensitive[1@n
359 [?2004l
360 Enabling conf deny-sensitive.
361 To activate the new configuration, you need to run:
362 systemctl reload apache2
363 [?2004hmallory@project1:~$ systemctl reload apache2
364 [?2004l
365 ==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
366 Authentication is required to reload 'apache2.service'.
367 Multiple identities can be used for authentication:
368 1. project1
369 2. mallory
370 Choose identity to authenticate as (1-2): 2
371 Password: Failed to reload apache2.service: Method call timed out
372
373 See system logs and 'systemctl status apache2.service' for details.
374
375 polkit-agent-helper-1: pam_authenticate failed: Authentication failure
376 [?2004hmallory@project1:~$ /!yQ=8:N2+8systemctl reload apache2
377 [?2004l
378 ==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
379 Authentication is required to reload 'apache2.service'.
380 Multiple identities can be used for authentication:
381 1. project1
382 2. mallory
383 Choose identity to authenticate as (1-2): 2
384 Password:
385 ==== AUTHENTICATION COMPLETE ====
386 [?2004hmallory@project1:~$ systemctl reload apache2
387 udo a2enconf deny-sensitive
388 [?2004l
389 Conf deny-sensitive already enabled
390 [?2004hmallory@project1:~$ sudo apache2 ctctl configtest
391 [?2004l
392 AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName
    ' directive globally to suppress this message
393 Syntax OK
394 [?2004hmallory@project1:~$ sudo systemctlsudo chown -R root:root /var/www/html
395 [?2004l
396 [?2004hmallory@project1:~$ sudo find /var/www/html -type d -exec chmod 755 {} \;
397 [?2004l
398 [?2004hmallory@project1:~$ sudo find /cvar/www/html -type f -exec chmod 644{} \;
399 [?2004l
400 chmod: missing operand after 644 /var/www/html/index-default.html
401 Try 'chmod --help' for more information.
402 chmod: missing operand after 644 /var/www/html/index.html
403 Try 'chmod --help' for more information.
404 chmod: missing operand after 644 /var/www/html/login/error.html
405 Try 'chmod --help' for more information.
406 chmod: missing operand after 644 /var/www/html/login/index.html
407 Try 'chmod --help' for more information.
408 chmod: missing operand after 644 /var/www/html/login/login.php
409 Try 'chmod --help' for more information.
410 chmod: missing operand after 644 /var/www/html/robots.txt
411 Try 'chmod --help' for more information.

```

```

412 [?2004hmallory@project1:~$ sudo find /var/www/html -type f -exec chmod 644{} \;[1@
413 [?2004l
414 [?2004hmallory@project1:~$ curl -I http://localhost/backup
415 [?2004l
416 HTTP/1.1 404 Not Found
417
418 Date: Thu, 12 Feb 2026 07:07:39 GMT
419
420 Server: Apache/2.4.58 (Ubuntu)
421
422 Content-Type: text/html; charset=iso-8859-1
423
424
425
426 [?2004hmallory@project1:~$ curl -I http://localhost/internal
427 [?2004l
428 HTTP/1.1 404 Not Found
429
430 Date: Thu, 12 Feb 2026 07:07:58 GMT
431
432 Server: Apache/2.4.58 (Ubuntu)
433
434 Content-Type: text/html; charset=iso-8859-1
435
436
437
438 [?2004hmallory@project1:~$ systemctl status apache2 --no-pager
439 [?2004l
440     apache2.service - The Apache HTTP Server
441     Loaded: loaded (/usr/lib/systemd/system/apache2.service; vendor preset: enabled)
442     Active: active (running) since Thu 2026-02-12 06:57:03 UTC; 11min ago
443     Docs: https://httpd.apache.org/docs/2.4/
444     Process: 1875 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
445     Process: 2005 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
446     Main PID: 1878 (apache2)
447     Tasks: 55 (limit: 4603)
448     Memory: 5.3M (peak: 7.1M)
449     CPU: 294ms
450     CGroup: /system.slice/apache2.service
451             └─1878 /usr/sbin/apache2 -k start
452             └─2010 /usr/sbin/apache2 -k start
453             └─2011 /usr/sbin/apache2 -k start
454 [?2004hmallory@project1:~$ exit
455 [?2004l
456 exit
457
458 Script done on 2026-02-12 07:08:25+00:00 [COMMAND_EXIT_CODE="0"]

```

A.2 Home Directories Remediation Transcript

Listing 13: Home Directories Permissions remediation transcript captured using script

```

1 Script started on 2026-02-12 15:29:56+00:00 [TERM="linux" TTY="/dev/tty1" COLUMNS="160" LINES="50"]
2 [?2004hmallory@project1:~/blue-work$ grep -Ri "password" /home/cd ..[Krm webserver.txt [4Pcd blue-work/grep -Ri "password" /
   home/[2Pfind /home/ f -perm -004[10Pls -ld /home/*
3 [?2004l
4 drwxr-x--- 7 alice users 4096 Feb 3 17:17 [0m[01;34m/home/alice[0m
5 drwxr-x--- 9 bob users 4096 Jan 30 19:03 [01;34m/home/bob[0m
6 drwxr-x--- 6 eve users 4096 Jan 30 14:18 [01;34m/home/eve[0m
7 drwxr-xr-x 7 ftpuser users 4096 Jan 30 19:13 [01;34m/home/ftpuser[0m
8 drwxr-x--- 10 mallory users 4096 Feb 12 03:09 [01;34m/home/mallory[0m
9 drwxr-x--- 4 project1 project1 4096 Feb 1 23:22 [01;34m/home/project1[0m
10 [?2004hmallory@project1:~/blue-work$ sudo chmod 7-- [K[K[K00 /home/f[Kt[Kftuser/
11 [?2004l
12 [sudo] password for mallory:
13 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/ftpuser/[C[C[C[K[K[K[K[K[K[K[Kalice/
14 [?2004l
15 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/alice/[K[K[K[K[K[K[Kbob/
16 [?2004l
17 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/bob/alice/[2Pbob/[K[K[K[KKeve/
18 [?2004l
19 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/eve/[K[K[K[Kmallory/
20 [?2004l
21 [?2004hmallory@project1:~/blue-work$ sudo chmod 700 /home/mallory/[4Peve/bob/alice/ftpuser/[15Pls -ld /home/*grep -Ri "
   password" /home/cd ..[Kgrep -Ri "password" /home/[12Pls -ld /home/*
22 [?2004l
23 drwx----- 7 alice users 4096 Feb 3 17:17 [0m[01;34m/home/alice[0m
24 drwx----- 9 bob users 4096 Jan 30 19:03 [01;34m/home/bob[0m
25 drwx----- 6 eve users 4096 Jan 30 14:18 [01;34m/home/eve[0m
26 drwx----- 7 ftpuser users 4096 Jan 30 19:13 [01;34m/home/ftpuser[0m
27 drwx----- 10 mallory users 4096 Feb 12 03:09 [01;34m/home/mallory[0m
28 drwxr-x--- 4 project1 project1 4096 Feb 1 23:22 [01;34m/home/project1[0m
29 [?2004hmallory@project1:~/blue-work$ find /hm[Kome/[K -type [K[K[K[K[K[K[K[K[K[KKe/[K =[K-type f -perm -004 2>/dev/null [K
30 [?2004l
31 /home/mallory/Downloads/readme.txt
32 /home/mallory/.sudo_as_admin_successful
33 /home/mallory/blue-work/homedirectories.txt
34 /home/mallory/audit/alice_review.txt
35 /home/mallory/.profile
36 /home/mallory/.cache/motd.legal-displayed
37 /home/mallory/.bashrc
38 /home/mallory/.bash_logout
39 [?2004hmallory@project1:~/blue-work$ find /home -type f -perm -004 2>/dev/null[C[1P 2>/dev/null2 2>/dev/null

```

```

40 [?20041
41 [?2004hmallory@project1:~/blue-work$ find /home -type f -perm -002 2>/dev/null[1P[1@d[C[C[C[C[C[C[C[C[C[C[1P 2>/dev/null5
    2>/dev/null
42 [?20041
43 /home
44 /home/mallory/Documents
45 /home/mallory/Downloads
46 /home/mallory/blue-work
47 /home/mallory/audit
48 /home/mallory/.config
49 /home/mallory/.ssh
50 /home/mallory/.local
51 [?2004hmallory@project1:~/blue-work$ ls -l [K[K[K[K-1 /home/mallory/*
52 [?20041
53 /home/mallory/audit:
54 total 4
55 -rw-r--r-- 1 mallory users 140 Jan 30 17:31 alice_review.txt
56
57 /home/mallory/blue-work:
58 total 0
59 -rw-r--r-- 1 mallory users 0 Feb 12 15:29 homedirectories.txt
60
61 /home/mallory/Documents:
62 total 24
63 -rw-r----- 1 mallory users 86 Jan 30 14:18 doc_1.txt
64 -rw-r----- 1 mallory users 86 Jan 30 14:18 doc_2.txt
65 -rw-r----- 1 mallory users 86 Jan 30 14:18 doc_3.txt
66 -rw-r----- 1 mallory users 86 Jan 30 14:18 doc_4.txt
67 -rw-r----- 1 mallory users 86 Jan 30 14:18 doc_5.txt
68 -rw-r----- 1 mallory users 86 Jan 30 14:18 doc_6.txt
69
70 /home/mallory/Downloads:
71 total 4
72 -rw-r--r-- 1 mallory users 50 Jan 30 14:18 readme.txt
73 [?2004hmallory@project1:~/blue-work$ exit
74 [?20041
75 exit
76
77 Script done on 2026-02-12 15:47:08+00:00 [COMMAND_EXIT_CODE="0"]

```